

Artificial neural networks

1 Problem overview

The Amazon rainforest is the largest tropical rainforest on Earth, renowned for its unparalleled biodiversity and vital role in regulating global climate patterns [1, 2]. It spans over 5.5 million km² and acts as a critical carbon sink by absorbing substantial amounts of atmospheric CO₂, thus mitigating global climate change [2]. However, recent decades have witnessed a concerning rise in temperatures across the Amazon basin, attributed primarily to climate change [3].

Within this vast rainforest (contoured in green in Figure 1), specific regions have experienced notably rapid increases in temperature, leading to more frequent and intense “hot events” [4]. These hot events threaten local ecosystems by increasing the risk of forest fires and have broader implications for global climate stability.

The occurrence and intensity of these hot events in the Amazon are influenced by large-scale climate drivers that modulate weather patterns worldwide [5]. In particular, four significant oceanic climate modes, El Niño Southern Oscillation (ENSO), Tropical South Atlantic (TSA), Tropical North Atlantic (TNA), and North Atlantic Oscillation (NAO), play crucial roles in determining temperature variations within the Amazon region. Each of these climate modes occurs in different oceanic regions surrounding the Amazon, as highlighted in Figure 1.

In this assignment, your task is to build a neural model using TensorFlow/Keras to predict monthly temperature and the occurrence of hot events in a specific Amazonian region, indicated in red in Figure 1. You will use monthly time-series data from 1982 to 2022, comprising temperature observations from the Amazon and indices representing the ENSO, TSA, TNA, and NAO climate modes.

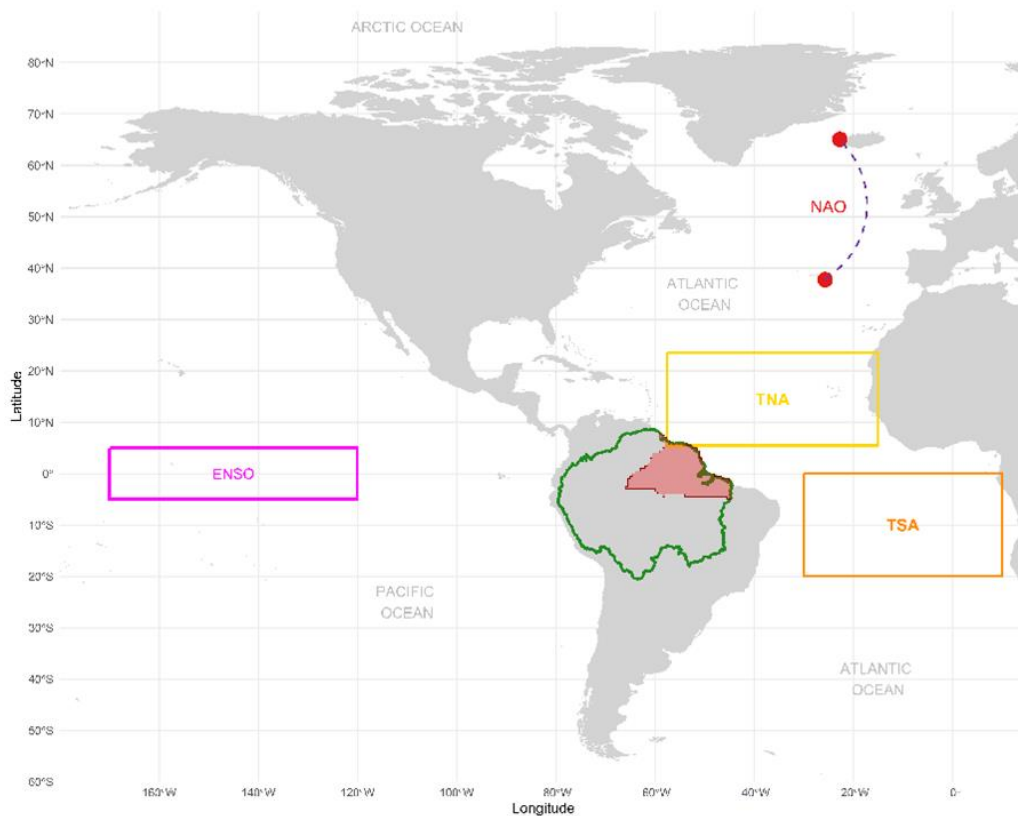


Figure 1: Map of the Amazon basin showing the study area (red) and the four surrounding climate-driver domains.

2 Provided Data

- Temperature time-series data and the values of the indices of climate modes ENSO, NAO, TSA and TNA as shown in Table 1 (Amazon_temperature_student.csv).
- Temperature thresholds provided for each month of the year (threshold.csv).

Table 1: Climate-mode indices: measurement, range and interpretation

Index	Measurement	Range	Interpretation
ENSO	Sea-surface temperature anomaly in Niño3.4 region	-3to3°C	+→ El Niño; -→ La Niña
NAO	Normalised sea-level pressure difference (Azores – Iceland)	-4to4	+→ stronger westerlies, milder winters; -→ reverse
TSA	SST anomaly in Tropical South Atlantic	-1to1°C	+→ warmer South Atlantic waters

TNA SST anomaly in Tropical North $-1\text{ to }1^\circ\text{C}$ $\rightarrow$ warmer North Atlantic waters
Atlantic

3 Objective

Your goal is to build neural network models for two distinct tasks:

- **Task A (Classification):** Predict whether a hot event occurs.
 - **Task B (Regression):** Predict the temperature.
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4 Task A: Classification (Hot event detection)

Data preparation

- Define a binary variable named Hot: A month is classified as Hot if the monthly temperature exceeds the provided threshold for that specific month.
 - Set Hot = 1 if the monthly temperature exceeds the monthly threshold.
 - Otherwise, set Hot = 0.
- Create a bar plot that summarises, for each year, the number of hot months.

Model development

- Randomly partition the dataset into training, validation, and test sets.
- Pre-processing:* Apply any necessary transformations to the training set, then apply the same transformations to the validation set. Keep a record of all applied transformations.
- Build a neural network classifier to predict the occurrence of hot events: define the architecture and hyper-parameters (loss function, optimiser, batch size, learning rate, number of epochs). It is recommended that the total number of trainable parameters obey

$$N_{\text{params}} < \frac{N_{\text{samples}}}{10},$$

i.e. keep the parameter count below one-tenth of the sample size.

- Create a plot showing the accuracy (y-axis) versus the number of epochs (x-axis) for both the training and validation sets.

Model evaluation

- Apply the same transformations to the test set as you did to the training and validation sets.

- (h) Use your model to predict the class Hot on the test set.
 - (i) Evaluate performance by plotting a confusion matrix. Note that your positive class is 1. (j) Calculate: Balanced Accuracy, True Negative Rate (Specificity), and True Positive Rate (Sensitivity).
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5 Task B: Regression (Temperature prediction)

For this task, directly predict temperature values rather than using the binary Hot variable.

Model development

- (k) Randomly split the dataset into training, validation, and test sets.
- (l) *Pre-processing*: Apply any necessary transformations to the *input features* of the training set, and replicate exactly the same transformations on the validation set. **Do not scale, normalise or otherwise transform the ground-truth (target) values.** Keep an explicit record of every transformation applied.
- (m) Build your model by defining its architecture and training hyper-parameters: loss function, optimiser, batch size, learning rate and number of epochs. It is recommended that the total number of trainable parameters be less than one-tenth of the number of samples.
- (n) Create a plot showing the values of the loss function (y-axis) versus the number of epochs (x-axis) for both the training and validation sets.

Model evaluation

- (o) Evaluate the regression model by comparing the true and predicted temperature values on the test set. Use the Pearson Correlation Coefficient (r) and the Mean Absolute Error (MAE) as evaluation metrics. The metrics are defined as

$$r = \frac{\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2}}, \quad \text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|,$$

where y_i is the true temperature, $\hat{y}_i$ is the predicted temperature, and n is the number of test samples.

Model development (year-wise split & target normalisation) [6]

- (p) Split the data by whole calendar years: each year must appear in one subset only. Use the same proportions as in part (k) for train/validation/test. Fit a *separate* scaler to the

training-set temperature targets and apply it to the validation and test targets; do *not* reuse the feature scaler. Retain the feature transformations specified in part (l) unchanged.

- (q) Work with the train/validation/test partitions—and the target-scaler—already established in part (p). Now apply every remaining feature-level pre-processing step specified in part (l) to those same splits.
- (r) Re-train the *same regression network and hyper-parameters defined in (m)* on the year-wise split (no new architecture).
- (s) Create a plot showing the values of the loss function (y-axis) versus the number of epochs (x-axis) for both the training and validation sets.

Model evaluation (Year-wise Split)

- (t) Evaluate the regression model by comparing the true and predicted temperature values on the test set. Use the Pearson Correlation Coefficient (r) and the Mean Absolute Error (MAE) as evaluation metrics.

6 Additional notes (apply to both tasks)

- You need to set a random seed to ensure that your results are reproducible.
- You must serialise both versions of every trained model *and* the corresponding feature-scaler objects feature scalers (and the separate target scalers from part (p) where applicable) for the random split **and** for the year-wise split.
- You cannot use year as a predictor. However, you may choose whether or not to include month as a predictor. If you include month, use cyclic encoding to represent it properly, ensuring your neural network correctly understands the continuity between December and January. To achieve cyclic encoding: Normalise the month to the range $[0, 2\pi]$ using: $\text{month normalised} = 2\pi \times \frac{(\text{month}-1)}{12}$.
- In the year-wise split, the years selected for training, validation, or testing do not need to be consecutive. For example, data from 1982 may belong to the training set, 1983 to the validation set, and 1984 to the test set.

References

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